THE READ-ONCE FACTORIAL-DIVISIBILITY CONJECTURE: ALGEBRAIC PROOFS THROUGH SIX INPUTS

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ABSTRACT. Let S be a set of n distinct integers. The read-once factorial-divisibility conjecture asks whether one can combine the elements of S, using each element exactly once and using only addition, subtraction, and multiplication, to obtain an element of $n!\mathbb{Z}$. The conjecture remains open in general. We prove it algebraically for every $1 \le n \le 6$. The proofs use congruences, signed residue classes, pigeonhole principles, and products of expressions on disjoint blocks; exhaustive enumeration is not used as proof. The crucial new ingredients for six inputs are a mixed-parity theorem stating that three odd inputs and one even input always yield a multiple of 60, and an auxiliary theorem producing a 120-multiple from four odd inputs coprime to 15. These close the parity patterns 3O3E and 4O2E. We also record current proved divisibility frontiers F_n, O_n, E_n for arbitrary, all-odd, and all-even inputs through $n = 6$, emphasize that incomparable moduli may require different expressions, and document the provenance and dominance reductions behind those frontiers.

1. INTRODUCTION

Arithmetic-expression construction usually asks whether prescribed numbers can be combined to reach a particular target. Here the target is a congruence class: from n distinct integers, every input must occur exactly once as a leaf of one binary expression tree, and the root value should lie in $n!\mathbb{Z}$. The read-once requirement prevents reuse in identities such as $(a-b)(a+b)$ and makes local congruence choices compete for leaves. This places the problem near arithmetic-expression construction [1] and arithmetic read-once formulas [3], while the repeated use of signed residue classes is reminiscent of zero-sum methods [4, 5]. The factorial target also connects naturally with classical questions about factorials and generalized factorials [2].

The central object of this paper is a conjecture, not a theorem for all inputs: the statement for general n remains open. The contribution here is to prove the conjecture by hand for the complete range $1 \le n \le 6$ and to organize the small cases into reusable divisibility profiles. A block need not have one preferred modulus. For example, three even integers universally yield both an 18-multiple and a 48-multiple, generally by different expressions. Neither modulus divides the other, so either can be the useful block in a larger construction.

Theorem 1.1 (Main theorem)**.** *For every integer n with $1 \le n \le 6$ and every set of n distinct integers, there exists a read-once expression using only $+, -, \times$ whose value belongs to $n!\mathbb{Z}$.*

The cases through five inputs are developed from small congruence theorems. The six-input case is divided by parity. Five of the seven parity patterns follow from earlier blocks. The remaining patterns are closed by

$$3\text{O1E} \to 60 \qquad \text{and} \qquad 4\text{O with all inputs coprime to } 15 \to 120.$$

The main text keeps the structural proof. Complete congruence branches are placed in the appendices so that no case is omitted while the main argument remains readable.

Date: August 2026.

2020 *Mathematics Subject Classification.* 11A07, 11B75, 05A05.

Key words and phrases. factorial divisibility, read-once arithmetic expression, congruences, signed residue classes, divisibility frontier, conjecture.

Preprint DOI: 10.5281/zenodo.21853500.

2. Definitions and the conjecture

Definition 2.1 (Read-once expression)**.** Let $S = \{a_1, \ldots, a_n\} \subset \mathbb{Z}$. A read-once arithmetic expression on S is a fully parenthesized expression formed from the binary operations $+, -$, and $\times$, in which each input occurs exactly once. Equivalently, it is a full binary tree whose leaves are in bijection with S and whose internal nodes are labelled by $+, -$, or $\times$. Parentheses determine the order of operations and do not count as additional operations.

For example,

$$(a_1 - a_2)(a_3 + a_4)$$

is a legal read-once expression on four inputs. By contrast, $a_1(a_2 - a_3) + a_1$ is not legal because a_1 occurs twice, and $(a_1 - a_2)^2$ is not legal because exponentiation is not an allowed operation and its expansion would reuse the inputs.

The value 0 is allowed, because $0 \in m\mathbb{Z}$ for every positive integer m. The formal statements concern sets of distinct integers; repeated-input cancellation is therefore not part of the problem.

Conjecture 2.2 (Read-once factorial-divisibility conjecture)**.** For every integer $n \geq 1$ and every set S of n distinct integers, there exists a read-once expression on S, using only $+, -, \times$, whose value lies in $n!\mathbb{Z}$.

Definition 2.3 (Guarantee notation)**.** Write $P_r(n; m)$ when every set of n distinct integers having exactly r odd elements admits a read-once expression divisible by m. Write $P(n; m)$ when the parity restriction is absent.

For compact parity notation, 3O1E $\to$ 60 means $P_3(4; 60)$. The read-once expression may depend on the actual input set; the conjecture does not require one fixed expression template to work for every set.

Definition 2.4 (Recorded source-pool divisibility frontiers)**.** For fixed n, let F_n denote the moduli maximal under divisibility among the universal guarantees explicitly included in the source pool described in Section E; let O_n and E_n be the corresponding sets for the all-odd and all-even source pools. A modulus m is removed when another modulus M in the same source pool satisfies $m \mid M$. In that situation we say that M *dominates* m, because the same expression that is a multiple of M is automatically a multiple of m.

For example, once 63 is proved for a class, the smaller guarantee 21 is not retained in its frontier because $21 \mid 63$. This dominance deletion is fundamentally different from combining guarantees obtained by different expressions: knowing separately that 6 and 8 are guaranteed does not imply that one expression is divisible by $\operatorname{lcm}(6, 8) = 24$. The recorded frontiers below summarize the displayed source pools; they are not optimality claims and are not asserted to be closed under every further case split that can be deduced from the proved parity theorems.

3. General construction principles

Lemma 3.1 (Block product)**.** *Suppose disjoint input blocks A and B admit read-once expressions E_A, E_B with $m \mid E_A$ and $q \mid E_B$. Then $E_A E_B$ is a read-once expression on $A \cup B$ divisible by mq.*

Proof. The two subexpressions use disjoint leaves, and their product uses every leaf in $A \cup B$ exactly once. □

Lemma 3.2 (Two-input factor 3)**.** *Any two integers can be combined into a multiple of* 3*.*

Proof. If one input is divisible by 3, use the product. Otherwise both are congruent to $\pm 1 \pmod 3$, so their sum or difference is divisible by 3. □

Lemma 3.3 (Signed pairing modulo $2n$)**.** *Among any n integers, either one input is divisible by n, or two inputs a, b satisfy $2n \mid a + b$ or $2n \mid a - b$.*

Proof. Modulo $2n$, a residue 0 or n represents an integer divisible by n. Otherwise the residues lie in the $n - 1$ signed classes $\{\pm 1\}, \ldots, \{\pm(n-1)\}$. Two of the n inputs occupy one class, and their suitable sum or difference is divisible by $2n$. □

Lemma 3.4 (Sign normalization and transport)**.** *In a congruence argument one may replace each input a_i by an independently chosen signed representative $\varepsilon_i a_i$, where $\varepsilon_i \in \{\pm 1\}$. More generally, let E be any read-once expression formed from the signed leaves $\varepsilon_i a_i$ using only the binary operations $+, -$, and $\times$. Then there is a legal read-once expression E_0 on the original leaves a_i, with the same underlying tree and using only $+, -$, and $\times$, such that $E = E_0$ or $E = -E_0$. Hence restoring the selected leaf signs does not change any divisibility conclusion.*

Proof. Proceed by induction on the expression tree. For a leaf εa, take the unsigned leaf a and keep ε only as an external sign marker. At a multiplication node, multiply the two external sign markers and multiply the two unsigned child expressions. At an addition or subtraction node, factor out the external sign marker of the left child; the relative sign of the right child determines whether that node is labelled $+$ or $-$. Thus every internal node remains one of the allowed binary operations, every original leaf is used exactly once, and only one overall sign remains at the root. An overall sign does not affect divisibility. □

Lemma 3.5 (Signed-class triangle)**.** *Suppose u, v, w lie in one signed residue class $\{\pm r\}$ modulo m, where $r \not\equiv 0 \pmod m$ and $2r \not\equiv 0 \pmod m$. For each pair, let $\sigma_m(u, v) \in \{\pm 1\}$ denote the unique sign satisfying $m \mid u + \sigma_m(u, v)v$. Then*

$$\sigma_m(u, v)\sigma_m(v, w)\sigma_m(w, u) = -1.$$

Proof. Write $u \equiv \varepsilon_u r$, $v \equiv \varepsilon_v r$ and $w \equiv \varepsilon_w r \pmod m$, with each $\varepsilon_i \in \{\pm 1\}$. The condition $2r \not\equiv 0 \pmod m$ makes the admissible sign unique, and $\sigma_m(u, v) = -\varepsilon_u \varepsilon_v$. Multiplying the three edge signs gives -1. □

The triangle relation will be used for $m = 3, 4, 5$. For $m = 3$, all nonzero residues form $\{\pm 1\}$; for $m = 4$, all odd residues form $\{\pm 1\}$; and for $m = 5$, the nonzero residues form $\{\pm 1\}$ and $\{\pm 2\}$.

4. Core blocks through five inputs

This section records the blocks needed in the six-input proof. Complete small-case derivations, including complementary moduli used to form the frontiers, appear in Section A.

Theorem 4.1 (Three-input 12 theorem)**.** *Every three integers admit a read-once expression divisible by 12; that is, $P(3; 12)$.*

Proof. See Theorem A.5. The proof separates the four parity patterns and uses only congruences modulo 3 and 4. □

Theorem 4.2 (Same-parity 18 and even-input 48)**.** *Any three integers of the same parity admit an expression divisible by 18, and any three even integers admit an expression divisible by 48:*

$$P_0(3; 18), \quad P_3(3; 18), \quad P_0(3; 48).$$

Proof. See Theorems A.6 and A.8. □

Theorem 4.3 (Four-input universal blocks)**.** *Every four integers admit expressions divisible by 24 and by 36:*

$$P(4; 24), \qquad P(4; 36).$$

Proof. For 24, apply Theorem 3.3 with $n = 4$. If one input is divisible by 4, multiply it by a 12-multiple from the remaining three. Otherwise a pair yields an 8-multiple, while the complementary pair yields a 3-multiple by Theorem 3.2.

For 36, if one input is divisible by 3, multiply it by a 12-multiple from the remaining three. Assume henceforth that no input is divisible by 3.

If the number of odd inputs is 0, 2, or 4, partition the four inputs into two same-parity pairs. In each pair choose a sum or difference divisible by 3. Because the two entries of the pair have the same parity, that same sum or difference is even, so each pair gives a 6-block. The two disjoint blocks therefore give 36.

If the parity pattern is 3O1E, the three odd inputs give an 18-multiple by Theorem 4.2, and multiplication by the remaining even input supplies an additional factor 2. Hence the full expression is divisible by 36.

Finally consider 1O3E. Let a be the odd input and x, y, z the even inputs. Among x, y, z, two are congruent modulo 4; relabel them x, y. Choose the sign for which $3 \mid x \pm y$. Since $x \equiv y \pmod 4$, both $x + y$ and $x - y$ are divisible by 4, so the same chosen block is divisible by 12. The remaining pair a, z can be combined into a multiple of 3 by Theorem 3.2. Multiplying the two disjoint blocks gives a multiple of 36.

These cases exhaust all parity patterns. □

Theorem 4.4 (Four-input parity blocks)**.** *The following guarantees hold:*

$$P_0(4;48),\ P_1(4;48),\ P_2(4;48),\ P_4(4;48),\ P_0(4;72),$$
$$P_1(4;40),\ P_2(4;40),\ P_3(4;60),\ P_4(4;60).$$

Proof. See Theorems A.9 to A.13 and section B. □

Theorem 4.5 (Five-input factorial target)**.** *Every five distinct integers admit a read-once expression divisible by* 120*; that is,* $P(5;120)$.

Proof. Apply Theorem 3.3 with $n = 5$. If one input is divisible by 5, multiply it by a 24-multiple from the remaining four. Otherwise a pair has a sum or difference divisible by 10, and the remaining three inputs yield a 12-multiple. Thus $10 \cdot 12 = 120$. □

Theorem 4.6 (Five odd and five even inputs)**.** *Five odd inputs always yield multiples of* 180 *and* 240*, and five even inputs always yield a multiple of* 480*:*

$$P_5(5;180), \qquad P_5(5;240), \qquad P_0(5;480).$$

Proof. See Theorems A.14 and A.15. □

Consequently Theorem 2.2 holds algebraically for $1 \le n \le 5$. The six-input proof requires one additional four-leaf block.

5. The mixed-parity modulus 60

Theorem 5.1 (Mixed-parity 60 theorem)**.** *Any three odd integers and one even integer admit a read-once expression divisible by* 60*:*

$$3\text{O}1\text{E} \to 60.$$

Proof. See Section B for the complete case analysis. □

Idea of proof. Let a, b, c be odd and e even. The proof first disposes of an input divisible by 5, then separates the case $3 \mid e$. Under $3 \nmid e$, the three odd inputs contain either zero, one, or at least two multiples of 3, giving an exhaustive partition. When no odd input is a multiple of 3, sign comparisons modulo 3, 4, and 5, together with the signed-class triangle relation, produce complementary blocks whose moduli multiply to 60. When exactly one odd input is a multiple of 3, sign normalization modulo 5 leaves exactly two possible signed-class positions for e; the explicit modulo-3, modulo-4, and modulo-5 calculations are given in Section B. Every simultaneous divisibility claim there uses one and the same choice of sign.

Theorem 5.2 (Four odd inputs coprime to 15)**.** *If four odd inputs are all coprime to* 15*, then they admit a read-once expression divisible by* 120*.*

Proof. Modulo 5, the four inputs occupy two nonzero signed classes. If the class-size distribution is $3+1$ or $4+0$, take three inputs u, v, w in one class. Were every edge to satisfy $\sigma_3 = -\sigma_5$, the products of the three edge signs modulo 3 and modulo 5 would be opposite. By Theorem 3.5 both products equal -1, a contradiction. Hence some pair satisfies $\sigma_3 = \sigma_5$, and its suitable sum or difference is divisible by 30. The remaining odd pair yields a multiple of 4.

If the distribution is $2 + 2$, write a, b in one class and c, d in the other. If either same-class pair, say a, b, satisfies $\sigma_3(a, b) = \sigma_5(a, b)$, then the same signed sum or difference is divisible

by 2, 3, and 5, so a, b form a 30-block; the remaining odd pair c, d forms a 4-block. Hence $30 \cdot 4 = 120$.

Assume therefore that both same-class pairs satisfy $\sigma_3 = -\sigma_5$. For the pair a, b, the binary sign $\sigma_4(a, b)$ must equal either $\sigma_3(a, b)$ or $\sigma_5(a, b)$. If $\sigma_4(a, b) = \sigma_3(a, b)$, then the same signed sum or difference of a, b is divisible by 12, while the pair c, d, using its modulo-5 sign, forms a 10-block; thus $12 \cdot 10 = 120$. If instead $\sigma_4(a, b) = \sigma_5(a, b)$, then a, b form a 20-block, while c, d, using their modulo-3 sign, form a 6-block; thus $20 \cdot 6 = 120$. □

6. Six inputs

Theorem 6.1 (Six-input factorial theorem)**.** *Every six distinct integers admit a read-once expression divisible by* $720 = 6!$.

Proof. We split by the number of odd inputs. The complete leaf-by-leaf constructions appear in Section C; the main exits are summarized below.

Parity pattern	Divisibility construction
0O6E	$18 \cdot 10 \cdot 4 = 720$
1O5E	$20 \cdot 36$, or $10 \cdot 2 \cdot 36$
2O4E	$10 \cdot 72$, $20 \cdot 36$, $18 \cdot 40$, or equivalent branches
3O3E	$6 \cdot 120$, or $12 \cdot 60$
4O2E	$6 \cdot 120$, $12 \cdot 60$, $20 \cdot 36$, or $120 \cdot 6$
5O1E	$60 \cdot 12$, or $20 \cdot 36$
6O0E	$18 \cdot 10 \cdot 4 = 720$

For 0O6E and 6O0E, one first constructs an 18-block, then a 10-block from two of the remaining four inputs, and finally a 4-block from the last pair. The parity of each selected pair supplies the factor 2 accompanying the modulo-9 or modulo-5 divisibility.

For 1O5E, an even multiple of 5 supplies 10; another even leaf supplies 2, and the remaining four leaves supply 36. If no even input is divisible by 5, two of the five even inputs collide in one of the four nonzero even signed classes modulo 20, producing a 20-block, and the remaining four leaves supply 36. The 5O1E argument is the odd analogue: an odd multiple of 5 is combined with a 12-block to produce 60, while in the absence of such an input a signed-class collision modulo 20 produces 20 and the complementary block produces 36.

For 2O4E, the proof first separates the presence of a multiple of 5. Otherwise it checks whether the odd pair shares a signed class modulo 5, whether an even pair collides modulo 20, and finally whether a multiple of 3 occurs. If none does, a pair of even inputs gives 18; the unused leaves give either $10 \cdot 4$ or the previously proved 2O2E $\to 40$. These branches are written leaf by leaf in Section C.5.

For 3O3E, if an even input is divisible by 3, it supplies 6 and the remaining five inputs supply 120. Otherwise the three even inputs are nonzero modulo 3. Two have the same residue modulo 4; choose the sign making their sum or difference divisible by 3. The same block is divisible by 4, hence by 12. The complementary 3O1E block supplies 60.

For 4O2E, first dispose of an even multiple of 3 by $6 \cdot 120$. If the two even inputs are congruent modulo 4, they yield a 12-block and the four odd inputs yield 60. Otherwise write the even inputs as $4 \mid x$ and $y \equiv 2 \pmod 4$. An odd multiple of 3 gives $12 \mid ax$ and leaves a 3O1E block of modulus 60. If no odd input is divisible by 3 but one is divisible by 5, then $20 \mid ax$ and the remaining four inputs give 36. In the final case all four odd inputs are coprime to 15, so Theorem 5.2 gives 120; the even pair gives 6.

The other five parity patterns are proved in Sections C.1 to C.5. The seven patterns exhaust all possibilities. □

Corollary 6.2. *Theorem 2.2 holds for every* $1 \le n \le 6$.

7. Recorded source-pool divisibility frontiers

The following table records the nondominated moduli in the explicit source pools used in this paper. Each entry is a proved existence guarantee for some expression; entries in the same row need not be attained simultaneously. The table is a provenance summary rather than an exhaustive closure of every parity-restricted consequence.

n	F_n (arbitrary)	O_n (all odd)	E_n (all even)
1	$\{1\}$	$\{1\}$	$\{2\}$
2	$\{2, 3\}$	$\{3, 4\}$	$\{6, 8\}$
3	$\{7, 8, 9, 10, 12\}$	$\{8, 10, 12, 14, 18\}$	$\{10, 14, 18, 48\}$
4	$\{10, 14, 16, 24, 36\}$	$\{14, 36, 48, 60\}$	$\{20, 28, 64, 72, 96\}$
5	$\{14, 16, 21, 27,$ $36, 120\}$	$\{32, 42, 54, 56,$ $72, 180, 240\}$	$\{84, 108, 112,$ $288, 384, 480\}$
6	$\{49, 56, 63, 64, 70,$ $81, 84, 96, 100, 108, 720\}$	$\{100, 112, 140, 168, 192,$ $196, 216, 252, 324, 720\}$	$\{100, 140, 196, 252, 324,$ $512, 672, 720, 864, 960, 2304\}$

The products and dominance deletions producing these rows are listed in Section E. Optimality is not claimed.

Beyond the source pool used in the factorial proof, we found further universal four-input guarantees. The additional constructions in Section F prove $P(4; m)$ for

$$m \in \{17, 19, 21, 22, 23, 25, 26, 27, 28, 30, 32, 33, 35, 40\}.$$

Together with the previously recorded 24- and 36-guarantees, the nondominated four-input universal moduli among the constructions displayed in this manuscript are

$$\{17, 19, 21, 22, 23, 24, 25, 26, 27, 28, 30, 32, 33, 35, 36, 40\}.$$

The additional four-input refinements are not propagated into the five- and six-input source-pool rows above. A parity-by-parity guide to the factorial constructions through six inputs is collected in Section D.

8. Guaranteed moduli through six inputs

To record one convenient universal modulus at each input size proved in this paper, we may take

$$L_1 = 1, \quad L_2 = 2, \quad L_3 = 12, \quad L_4 = 24, \quad L_5 = 120, \quad L_6 = 720.$$

Each L_n is a separately proved guarantee; no optimality is asserted, and no recurrence beyond six inputs is pursued here. In particular, the general factorial-divisibility conjecture remains open beyond the range covered by Theorem 1.1.

9. Exact residue dynamic programming as a checking tool

For a fixed modulus M and fixed inputs, exact read-once reachability can be computed over residues. For every nonempty subset T of the input indices, let $\mathcal{R}(T)$ be the residues modulo M attained by expressions using exactly the leaves in T. Initialize singleton sets by the input residues. For every bipartition $T = A \sqcup B$, combine $x \in \mathcal{R}(A)$ and $y \in \mathcal{R}(B)$ by

$$x + y, \quad x - y, \quad y - x, \quad xy \pmod{M}.$$

Induction on $|T|$ shows that this recurrence is complete.

The algorithm is not used as proof here. It serves only as an independent consistency check of delicate blocks. The algebraic appendices, not finite search, establish the stated theorems.

10. Conclusion

The read-once factorial-divisibility statement remains a conjecture for general n. This paper proves the complete range through six inputs. The decisive step is not exhaustive residue enumeration, but reusable congruence blocks with complementary prime-power profiles, especially 3O1E $\to$ 60 and the four-odd coprime-to-15 modulus 120. Together with the earlier three-, four-, and five-input results, these close all seven parity patterns for $n = 6$. The first unresolved finite case is $n = 7$.

Generative AI disclosure

OpenAI ChatGPT (GPT-5.6) was used for exploratory mathematical discussion, organization and revision of proof arguments, preparation and review of auxiliary consistency-checking code, and English-language editing. The tool assisted in examining possible proof structures, identifying arguments requiring clarification, and improving the presentation of the manuscript. It was not used as a substitute for mathematical proof. The author reviewed all definitions, theorem statements, proofs, computations, references, and conclusions and assumes full responsibility for the originality, accuracy, and integrity of the manuscript.

Declarations

Funding. No external funding was received for this work.

Competing interests. The author declares no competing interests.

Data and code availability. No external dataset is associated with this work. The LaTeX source accompanies the preprint record. The algebraic proofs are self-contained. A supplementary Python script supplied with the source independently reconstructs the residual signed-class lists used in the 17/19, 23, and 35 arguments and checks their displayed congruence certificates; it serves only as a consistency check and is not required for the proofs.

Appendix A. Complete small-block proofs through five inputs

This appendix supplies the detailed algebraic constructions used before the six-input theorem. Every displayed product combines disjoint blocks, and all signs refer to a choice of sum or difference.

A.1. **One and two inputs.** For one input, divisibility by 1 is immediate. For two arbitrary inputs a, b, if they have the same parity then $a - b$ is even; otherwise ab is even. This proves $P(2; 2)$. If one input is divisible by 3, their product is divisible by 3; otherwise both are nonzero modulo 3 and a suitable $a \pm b$ is divisible by 3. Hence $P(2; 3)$.

If a, b are odd, one of $a + b, a - b$ is divisible by 4. If a, b are even, then the preceding factor-3 construction is automatically even, proving a 6-multiple. For a multiple of 8, if one input is divisible by 4 use the product; otherwise both are 2 (mod 4), and according as their residues modulo 8 agree or are opposite, their difference or sum is divisible by 8.

A.2. **Three arbitrary inputs.**

Theorem A.1. *Every three integers admit an expression divisible by* 7*.*

Proof. If one input is divisible by 7, multiply it by any legal expression on the other two leaves. Otherwise the residues lie in the three nonzero signed classes $\{\pm 1\}, \{\pm 2\}, \{\pm 3\}$ modulo 7. A collision gives a sum or difference divisible by 7; multiply that block by the remaining third leaf. If all three classes occur, relabel and choose signed representatives so that $a \equiv 1$, $b \equiv 2$, $c \equiv 3$ (mod 7). Since $bc \equiv -1$ (mod 7), one of the legal expressions $bc + a, bc - a, a - bc$ is divisible by 7 after restoring the selected signs. □

Theorem A.2. *Every three integers admit an expression divisible by* 8*.*

Proof. If all three are even, use the product. If exactly two, x, y, are even and z is odd, use the product when one even input is divisible by 4. Otherwise $x, y \equiv 2 \pmod 4$, and one of $x+y, x-y$ is divisible by 8; multiply it by z.

If exactly one input e is even and a, b are odd, then when $4 \mid e$ use $e(a+b)$, while when $e \equiv 2 \pmod 4$ choose $a \pm b$ divisible by 4 and use $e(a \pm b)$. Finally, for three odd inputs, the residues occupy the two signed classes $\{\pm 1\}, \{\pm 3\}$ modulo 8; a colliding pair has a sum or difference divisible by 8, which is multiplied by the third input. □

Theorem A.3. *Every three integers admit an expression divisible by* 9*.*

Proof. If one input is divisible by 9, use the product. If at least two inputs are divisible by 3 but none by 9, those two are in the signed class $\{\pm 3\} \pmod 9$, so their suitable sum or difference is divisible by 9; multiply that block by the remaining third leaf. If exactly one input c is divisible by 3, the other two are nonzero modulo 3 and a suitable $a \pm b$ is divisible by 3; hence $9 \mid c(a \pm b)$.

It remains that no input is divisible by 3. The nonzero signed classes modulo 9 are

$$\{\pm 1\}, \quad \{\pm 2\}, \quad \{\pm 4\}.$$

A collision gives a 9-multiple pair, which is multiplied by the remaining third leaf. If all three classes occur, normalize to $a \equiv 1$, $b \equiv 2$, $c \equiv 4 \pmod 9$. Then $bc \equiv -1 \pmod 9$, so a suitable $bc \pm a$ is divisible by 9. □

Theorem A.4. *Every three integers admit an expression divisible by* 10*.*

Proof. If one input, say a, is divisible by 5 and at least one of the three inputs is even, the full product abc is divisible by 10. If all three inputs are odd, use $a(b+c)$, which is divisible by 10 because $b+c$ is even.

Assume no input is divisible by 5. Two inputs lie in one of the signed classes $\{\pm 1\}, \{\pm 2\}$ modulo 5; call them a, b. Choose a sign so that $5 \mid a \pm b$. If a, b have the same parity, this block is divisible by 10 and is multiplied by the third input c. If they have opposite parity and c is even, then $c(a \pm b)$ is divisible by 10.

The only remaining parity pattern has a, b of opposite parity and c odd. If the two odd inputs lie in the same signed class modulo 5, their suitable sum or difference is divisible by 10 and is multiplied by the remaining even input. Otherwise, after sign normalization, the two inputs in one signed class contribute r, r and the odd input in the other class contributes $-2r$ modulo 5. Their signed three-term sum is divisible by 5 and is even (odd + even − odd), hence divisible by 10. □

Theorem A.5 (Three-input 12 theorem)**.** *Every three integers admit an expression divisible by* 12*.*

Proof. If all three are even and one is divisible by 3, their product is divisible by 24. If none is divisible by 3, choose $3 \mid a \pm b$; the block $a \pm b$ together with the remaining even factor supplies a factor 4.

Suppose exactly two inputs x, y are even and z is odd. If any input is divisible by 3, the product contains $4 \cdot 3$. Otherwise, if one even input, say x, is divisible by 4, choose $3 \mid y \pm z$ and use $x(y \pm z)$. If neither even input is divisible by 4, then $x \equiv y \equiv 2 \pmod 4$; choose $3 \mid x \pm y$. Either sign is divisible by 4, so $(x \pm y)z$ is divisible by 12.

Suppose exactly one input e is even and a, b are odd. If $4 \mid e$, then either the product already contains a factor 3, or a suitable $a \pm b$ is divisible by 3; in either case a 12-multiple results. Now let $e \equiv 2 \pmod 4$. If $3 \mid e$, choose $a \pm b$ divisible by 4. If neither odd input is divisible by 3, choose $a \pm b$ divisible by 3; this block is even and e supplies the second factor 2. If both odd inputs are divisible by 3, choose $a \pm b$ divisible by 4; it is also divisible by 3.

In the remaining subcase exactly one odd input, say a, is divisible by 3, while b, e are nonzero modulo 3. Choose the sign on b so that $e \pm b \equiv 0 \pmod 3$. The number $e \pm b$ is odd. Since a is odd, one of $(e \pm b) + a$ and $(e \pm b) - a$ is divisible by 4. Changing the sign of a does not affect divisibility modulo 3, so the resulting signed sum is divisible by 12.

Finally suppose all three inputs are odd. If one is divisible by 3, use it as a factor and combine the other two into a multiple of 4. If none is divisible by 3, the residues lie in the two signed

classes $\{\pm 1\}, \{\pm 5\}$ modulo 12; two inputs have a sum or difference divisible by 12, which is multiplied by the third input. □

A.3. **Three-input parity improvements.**

Theorem A.6 (Same-parity 18 theorem)**.** *Any three integers of the same parity admit an expression divisible by* 18*.*

Proof. Work modulo 9. If one input c is divisible by 9, then $c(a+b)$ is divisible by 18 because $a+b$ is even. If at least two inputs are divisible by 3 but none by 9, those two are congruent to ± 3 modulo 9; a suitable sum or difference is divisible by 9 and, by same parity, by 2. Multiply this 18-multiple block by the remaining third leaf.

If exactly one input c is divisible by 3, the other two are nonzero modulo 3, so choose $3 \mid a \pm b$. Then $9 \mid c(a \pm b)$, and $a \pm b$ is even. If no input is divisible by 3, use the signed classes $\{\pm 1\}, \{\pm 2\}, \{\pm 4\}$ modulo 9. A collision gives an even 9-multiple, which is multiplied by the remaining third leaf. If all three classes occur, normalize to $a \equiv 1, b \equiv 2, c \equiv 4 \pmod 9$; then $bc \equiv -1$, so $bc \pm a$ can be made divisible by 9. Since all inputs have the same parity, bc and a have the same parity, so the expression is even. □

Lemma A.7. *For any integers x, y, z, either $6 \mid xyz$, or after relabelling there is a sign such that $12 \mid (x \pm y)z$.*

Proof. Assume $6 \nmid xyz$. If xyz is odd and one input is divisible by 3, use it as z and choose $x \pm y$ divisible by 4. If none is divisible by 3, two lie in one of the signed classes $\{\pm 1\}, \{\pm 5\}$ modulo 12, and their suitable sum or difference is divisible by 12.

If xyz is even, no input is divisible by 3. If one input is divisible by 4, use it as z and choose $x \pm y$ divisible by 3. Otherwise every even input is $2 \pmod 4$. With exactly one even input, use it as z and choose the two odd inputs so that $3 \mid x \pm y$; the latter block is also even. With at least two even inputs, choose two as x, y and make $3 \mid x \pm y$; their sum or difference is automatically divisible by 4. □

Theorem A.8 (Three even inputs give 48)**.** *Any three even integers admit an expression divisible by* 48*.*

Proof. Write the inputs as $2x, 2y, 2z$. If $6 \mid xyz$, their product is divisible by 48. Otherwise Theorem A.7 gives $12 \mid (x \pm y)z$, and

$$(2x \pm 2y)(2z) = 4(x \pm y)z$$

is divisible by 48. □

The proof of Theorem A.1 also improves to 14 when the three inputs have one parity. If one input is divisible by 7, multiply it by the sum of the other two same-parity leaves, which is even. In the collision branch the 7-divisible sum or difference of the colliding same-parity pair is even and is multiplied by the third leaf. In the three-class branch, the product of two same-parity inputs and the remaining input have the same parity, so the constructed $bc \pm a$ expression is even.

A.4. **Four-input universal and parity blocks.** Besides the 24 and 36 theorems proved in the main text, every four-input set yields 10, 14, and 16. The modulus 10 follows by applying Theorem A.4 to three leaves and multiplying by the fourth. For 14, if an even input is present, reserve it and construct a 7-multiple from the other three; if all four are odd, construct a 14-multiple from three odd leaves and multiply by the last odd leaf.

For 16, classify by the number of odd inputs. Four even inputs have product divisible by 16. With one odd input, the three even inputs yield 48. With two odd and two even inputs, the odd pair yields 4 and the product of the even pair yields 4. With three odd inputs, those three yield an 8-multiple and the remaining even leaf supplies another factor 2. With four odd inputs, split into two pairs, each yielding 4.

Theorem A.9 (Four-input 48 refinements)**.** *A* 48*-multiple is guaranteed for four inputs containing exactly* $0, 1, 2$*, or* 4 *odd integers.*

Proof. For four even inputs, first choose an 8-block. If one input is divisible by 4, multiply it by any other even input; their product is divisible by 8. If no input is divisible by 4, then all four inputs are congruent to 2 modulo 4, hence each is congruent to 2 or 6 modulo 8. They all lie in the single signed class $\{\pm 2\}$ modulo 8, so any selected pair has a suitable sum or difference divisible by 8. The two remaining even inputs can be combined into a multiple of 3 by Theorem 3.2; that expression is automatically even, hence is a 6-block. The two disjoint blocks give $8 \cdot 6 = 48$. With exactly one odd input, the three even leaves yield 48 and the odd leaf is multiplied afterward.

For two odd inputs a, b and two even inputs x, y, if one even input is divisible by 4, reserve it and apply the three-input 12 theorem to the other leaves. Otherwise $x \equiv y \equiv 2 \pmod 4$. If one even input is divisible by 3, then xy is divisible by 12, while $a \pm b$ can be made divisible by 4. If neither even input is divisible by 3, choose $x \pm y$ divisible by 3; the same block is divisible by 12, and again choose $a \pm b$ divisible by 4.

For four odd inputs, the residues modulo 12 occupy the signed classes $\{\pm 1\}, \{\pm 3\}, \{\pm 5\}$. Two inputs yield a 12-multiple, while the remaining odd pair yields a 4-multiple. □

Theorem A.10 (Four even inputs give 72)**.** *Any four even integers admit an expression divisible by* 72*.*

Proof. If one input is divisible by 3, it supplies 6 and the remaining three even inputs supply 12. Otherwise all four inputs are nonzero modulo 3. Modulo 12, even nonmultiples of 3 occupy the signed classes $\{\pm 2\}, \{\pm 4\}$, so two yield a 12-multiple. The remaining even pair yields a multiple of 3, automatically even, hence a 6-multiple. □

Theorem A.11 (Four odd inputs give 60)**.** *Any four odd integers admit an expression divisible by* 60*.*

Proof. If one input is divisible by 5, multiply it by a 12-multiple from the other three odd inputs. Suppose no input is divisible by 5. If two inputs lie in one signed class modulo 20, their suitable sum or difference is divisible by 20, while the remaining pair yields a multiple of 3.

It remains that the four inputs occupy the four signed classes $\{\pm 1\}, \{\pm 3\}, \{\pm 7\}, \{\pm 9\}$ modulo 20. Normalize and label them a, b, c, d in that order. The pair b, c has a suitable sum or difference divisible by 10. If a, d are both divisible by 3 or both nonzero modulo 3, they yield a 6-multiple, completing $6 \cdot 10$.

If exactly one of a, d is divisible by 3, suppose first $3 \mid a$. Modulo 20, the product of the classes of b and d is the class of c, so a suitable $bd \pm c$ is divisible by 20; multiplication by a gives 60. If $3 \mid d$, similarly the class of bc equals that of a, so a suitable $bc \pm a$ is divisible by 20 and multiplication by d completes the proof. □

Theorem A.12 (One odd and three even inputs give 40)**.** *Any one odd and three even inputs admit an expression divisible by* 40*.*

Proof. Let the odd input be a and the even inputs x, y, z. If $5 \mid a$, multiply a by an 8-multiple from x, y, z. If an even input is divisible by 5, it supplies 10 and the other three inputs supply 8.

Assume no input is divisible by 5. If two even inputs are in the same signed class modulo 5 and have the same residue modulo 4, their suitable sum or difference is divisible by 20; the remaining odd-even product is even.

Otherwise the three even inputs have signed-class distribution $2 + 1$ modulo 5 (the $3 + 0$ distribution would contain a pair with the same residue modulo 4). Let x, y be the majority-class pair and z the singleton. Then x, y have different residues modulo 4. If $4 \mid z$, the pair x, y yields a 10-multiple and az yields a 4-multiple. Finally let $z \equiv 2 \pmod 4$. In the majority pair one input is $0 \pmod 4$ and the other $2 \pmod 4$. Pair the odd input a with the $2 \pmod 4$ even input in the same modulo-5 signed class. If a lies in the majority class, use the $2 \pmod 4$ member of the majority pair. If a does not lie in the majority class, then, because the nonzero residues modulo 5 form only two signed classes, a must lie in the singleton class; hence a and z lie in the same class. Their suitable sum or difference is divisible by 5. The two unused even inputs have product divisible by 8. Thus the product is divisible by 40. □

Theorem A.13 (Two odd and two even inputs give 40)**.** *Any two odd and two even inputs admit an expression divisible by* 40*.*

Proof. Let the odd inputs be p, q and the even inputs x, y. If one even input is divisible by 4, multiply it by a 10-multiple from the remaining three leaves. Hence assume $x \equiv y \equiv 2 \pmod 4$.

If some input is divisible by 5, reserve its factor 5 (or 10 for an even input) and use the three-input 8 theorem on the other leaves. Now assume all four inputs are nonzero modulo 5.

If p, q occupy one signed class modulo 5, their suitable sum or difference is divisible by 10, and xy is divisible by 4. If x, y occupy one signed class modulo 5, their suitable sum or difference is divisible by 5 and by 4, while the odd pair yields 4.

It remains that each parity pair occupies the two different signed classes modulo 5. Relabel so that p, x occupy one class and q, y the other. Consider

$$E_1 = px \pm qy, \qquad E_2 = py \pm qx.$$

Choose signed representatives modulo 5 so that

$$p \equiv \varepsilon_p r, \qquad x \equiv \varepsilon_x r, \qquad q \equiv 2\varepsilon_q r, \qquad y \equiv 2\varepsilon_y r,$$

where each ε is $+1$ or -1. Put $\delta = \varepsilon_p\varepsilon_q\varepsilon_x\varepsilon_y$. Direct multiplication of these congruences shows that E_1 is divisible by 5 when its sign is δ, whereas E_2 is divisible by 5 when its sign is $-\delta$. Thus the modulo-5 signs required by E_1 and E_2 are opposite.

Now write $x = 2X$ and $y = 2Y$, where X, Y are odd. For a sign $s \in \{\pm 1\}$,

$$E_1 = 2(pX + sqY), \qquad E_2 = 2(pY + sqX).$$

Divisibility by 8 is therefore equivalent to divisibility by 4 of the corresponding parenthesized odd sum or difference. The two pairs of odd terms satisfy

$$(pX)(qY) = (pY)(qX) = pqXY.$$

For two odd integers modulo 4, the sign that makes their sum or difference zero modulo 4 depends only on whether their residues are equal or opposite. The displayed product identity shows that the pair pX, qY is equal or opposite modulo 4 in exactly the same way as the pair pY, qX. Hence E_1 and E_2 require the same sign for divisibility by 8.

The modulo-5 signs for E_1 and E_2 are opposite, whereas their modulo-8 signs are the same. Therefore, for exactly one of E_1, E_2, the same choice of sign gives divisibility by both 5 and 8. That legal read-once expression is divisible by 40. □

The theorem 3O1E $\to$ 60 is proved separately in Section B. The simpler 3O1E $\to$ 20 follows by constructing a 10-multiple from the three odd inputs and multiplying by the even input.

A.5. **Five-input parity theorems.**

Theorem A.14 (Five odd inputs)**.** *Five odd inputs admit expressions divisible by* 180 *and by* 240*.*

Proof. Suppose first that no input is divisible by 5. The five inputs occupy the four nonzero odd signed classes modulo 20, so a pair yields a 20-multiple. The remaining three odd inputs yield an 18-multiple, giving 360, hence 180; using instead their universal 12-multiple gives 240.

If one input a is divisible by 5, among the other four odd inputs choose a pair b, c whose sum or difference is divisible by 12. Then $a(b \pm c)$ is divisible by 60. The remaining odd pair can be combined into a multiple of 3, giving 180, or into a multiple of 4, giving 240. □

Theorem A.15 (Five even inputs)**.** *Five even inputs admit an expression divisible by* 480*.*

Proof. If one input is divisible by 5, it is divisible by 10; the remaining four even inputs yield 48. Otherwise the five inputs occupy the two nonzero signed classes modulo 5, so two have a suitable sum or difference divisible by 5 and by 2, hence by 10. The remaining three even inputs yield 48. In both branches the disjoint blocks give 480. □

Appendix B. Complete proof of the mixed-parity 60 theorem

Let a, b, c be odd and e even. We prove that these four leaves can be used exactly once to construct a multiple of 60. The top-level partition is exhaustive: Case A handles an input divisible by 5; after excluding that case, Case B handles $3 \mid e$; under $3 \nmid e$, Cases D, E, and C handle respectively zero, exactly one, and at least two odd multiples of 3. Within every case, a statement that one expression is divisible by two moduli always refers to one and the same choice of sign. Signs belonging to different disjoint blocks may be chosen independently; within any single block, however, the same sign is used simultaneously for all stated congruence conditions.

Case A: one input is divisible by 5. If an odd input, say a, is divisible by 5, multiply it by a 12-multiple constructed from b, c, e. If $5 \mid e$, then $10 \mid e$ and the remaining three odd inputs yield a 12-multiple. Hence this case is complete. From now on all four inputs are nonzero modulo 5.

Case B: $3 \mid e$. Two of the three odd inputs lie in one signed class modulo 5, so their suitable sum or difference is divisible by 5 and, because both are odd, by 2; this is a 10-block. The remaining odd input multiplied by e is divisible by 6. Hence $10 \cdot 6 = 60$. From now on $3 \nmid e$.

Case C: at least two odd inputs are divisible by 3. If two such odd inputs lie in one signed class modulo 5, their suitable sum or difference is divisible by 2, 3, and 5, hence by 30; the remaining odd-even product is even.

Otherwise exactly two odd inputs are divisible by 3 and they occupy the two different signed classes modulo 5. The third odd input lies in the same class as one of them. That same-class odd pair gives a 10-block, while the unused multiple-of-3 odd input multiplied by e gives a 6-block. Thus this case is complete.

Case D: no odd input is divisible by 3. Now all four inputs are coprime to 15. Choose two odd inputs a, b in one signed class modulo 5. We distinguish the class positions of c, e.

D1. The remaining inputs c, e occupy one signed class modulo 5. For the pair a, b, compare the three signs $\sigma_3(a,b), \sigma_4(a,b), \sigma_5(a,b)$. At least two are equal. If $\sigma_3 = \sigma_5$, the pair yields 30 and ce yields 2. If $\sigma_4 = \sigma_5$, the pair yields 20, while $c \pm e$ can be made divisible by 3. If $\sigma_3 = \sigma_4$, the pair yields 12, while the same-class pair c, e yields 5. Each branch gives 60.

D2. The three odd inputs a, b, c occupy one signed class modulo 5, and e occupies the other. If an odd pair has $\sigma_3 = \sigma_5$, then the same signed sum or difference is divisible by 2, 3, and 5, hence forms a 30-block; the remaining odd-even product supplies 2. If an odd pair has $\sigma_4 = \sigma_5$, then the same signed sum or difference forms a 20-block, while the remaining odd-even pair can be combined into a 3-block.

Suppose neither exit occurs for any odd pair. Since each admissible sign is binary, every edge of the odd triangle must satisfy

$$\sigma_3 = \sigma_4 = -\sigma_5.$$

The signed-class triangle lemma says that the product of the three σ_3 edge signs is -1 and that the product of the three σ_5 edge signs is also -1. On the other hand, multiplying the three relations $\sigma_5 = -\sigma_3$ shows that the σ_5 edge-sign product is the negative of the σ_3 edge-sign product, because three negative factors occur. The σ_3 product is -1, so this would make the σ_5 product $+1$, a contradiction. Hence one of the two preceding exits must occur.

D3. The inputs a, b, e occupy one signed class modulo 5, and c the other. If $\sigma_3(a,b) = \sigma_5(a,b)$, then the same signed sum or difference of a, b forms a 30-block and the complementary product ce supplies 2. If $\sigma_4(a,b) = \sigma_5(a,b)$, then the same signed sum or difference of a, b forms a 20-block and the complementary pair c, e can be combined into a 3-block. Otherwise put

$$\rho(u,v) = \sigma_3(u,v)\sigma_5(u,v).$$

On the same-class triangle a, b, e, the two triangle identities give

$$\rho(a,b)\rho(b,e)\rho(e,a) = 1.$$

Since $\rho(a,b) = -1$, exactly one of the other two factors equals $+1$. Thus either a, e or b, e has one sum or difference divisible by both 3 and 5, giving a 15-block. The complementary odd pair yields 4.

This completes the case in which no odd input is divisible by 3.

Case E: exactly one odd input is divisible by 3. Let $3 \mid a$ and $3 \nmid bce$. If b, c occupy one signed class modulo 5, they yield a 10-block and ae yields a 6-block. Hence assume b, c occupy different signed classes. Relabel so that a, b occupy one class and c the other. There are two positions for e.

E1. The even input is in the class of a, b. By sign normalization, choose representatives satisfying

$$a \equiv b \equiv e \equiv r, \qquad c \equiv -2r \pmod 5.$$

Consequently

$$5 \mid a-b, \quad 5 \mid b-e, \quad 5 \mid a-e,$$
$$5 \mid a+b+c, \quad 5 \mid a+c+e, \quad 5 \mid b+c+e.$$

If $4 \mid a-b$, then $a-b$ is a 20-block and c, e can be combined into a 3-block. If $3 \mid b-e$, then $b-e$ is a 15-block and a, c can be combined into a 4-block. If there exists a single sign $s \in \{\pm 1\}$ such that

$$3 \mid b+sc \qquad \text{and} \qquad 4 \mid b+sc,$$

then $b+sc$ is a 12-block and $a-e$ is a 5-block.

Assume none of these three exits occurs. Since a, b are odd, exactly one of $a-b$ and $a+b$ is divisible by 4, so $4 \nmid a-b$ implies $4 \mid a+b$. Since b, e are both nonzero modulo 3, exactly one of $b-e$ and $b+e$ is divisible by 3, so $3 \nmid b-e$ implies $3 \mid b+e$. Finally, for the odd pair b, c, the modulo-3 sign and the modulo-4 sign are each unique. The absence of a single sign producing a 12-block means that these two signs are opposite. Thus

$$4 \mid a+b, \qquad 3 \mid b+e,$$

and the signs that make $b \pm c$ divisible by 3 and by 4 are opposite.

First let $e \equiv 0 \pmod 4$ and put $T = b+c+e$. We know $5 \mid T$. If $4 \mid T$, then aT is divisible by $3 \cdot 20 = 60$. If $4 \nmid T$, then $b+c \equiv 2 \pmod 4$, so $4 \mid b-c$. Because the modulo-3 and modulo-4 signs for $b \pm c$ are opposite, $3 \mid b+c$. Hence $15 \mid a+b+c$, and multiplication by the unused input e, divisible by 4, gives 60.

Now let $e \equiv 2 \pmod 4$, again with $T = b+c+e$. If $4 \mid T$, the product aT is divisible by 60. Otherwise $4 \mid b+c$, so the opposite-sign condition gives $3 \mid b-c$. Together with $3 \mid b+e$, this yields $3 \mid c+e$. Put

$$U = a+c+e.$$

Modulo 5, $U \equiv r - 2r + r \equiv 0$. Modulo 3, both a and $c+e$ are zero. Finally $4 \mid a+b$ and $4 \mid b+c$ imply $a \equiv c \equiv -b \pmod 4$, and therefore

$$U \equiv -2b + 2 \equiv 0 \pmod 4$$

because b is odd. Thus $60 \mid U$, and bU uses the final leaf.

E2. The even input is in the class of c. Normalize signs so that

$$a \equiv b \equiv r, \qquad c \equiv e \equiv -2r \pmod 5.$$

Then

$$5 \mid a-b, \quad 5 \mid c-e, \quad 5 \mid b-c-e, \quad 5 \mid a-c-e.$$

If $4 \mid a-b$, combine the resulting 20-block with a 3-block from c, e. If $3 \mid c-e$, combine the resulting 15-block with a 4-block from a, b. Assume neither exit occurs. Since a, b are odd, exactly one of $a-b$ and $a+b$ is divisible by 4; hence $4 \nmid a-b$ implies $4 \mid a+b$. Since c, e are nonzero modulo 3, exactly one of $c-e$ and $c+e$ is divisible by 3; hence $3 \nmid c-e$ implies $3 \mid c+e$. Thus

$$4 \mid a+b, \qquad 3 \mid c+e.$$

Put $T = b-c-e$. If $4 \mid T$, then aT is divisible by 60. Otherwise T is even but not divisible by 4, so $T \equiv 2 \pmod 4$. Put $U = a-c-e$. We have $5 \mid U$ from the normalization and $3 \mid U$ from $3 \mid a$ and $3 \mid c+e$. Moreover $a \equiv -b \pmod 4$ and $c+e \equiv b-2 \pmod 4$, so

$$U \equiv -b-(b-2) = 2-2b \equiv 0 \pmod 4$$

because b is odd. Thus $60 \mid U$, and bU uses the remaining input.

Cases A–E exhaust all possibilities. Therefore 3O1E $\to$ 60.

Appendix C. Complete proof of the six-input theorem

This appendix gives the full parity-by-parity construction for Theorem 6.1. In every product below, the displayed factors are built from disjoint leaves.

C.1. **Six even inputs.** First construct an 18-block. If one input is divisible by 9, multiply it by another even input. Otherwise six residues occupy the four nonzero signed classes $\{\pm1\}, \{\pm2\}, \{\pm3\}, \{\pm4\}$ modulo 9, so two collide; their suitable sum or difference is divisible by 9 and by 2.

Among the remaining four even inputs, construct a 10-block. If one is divisible by 5, multiply it by another even input. Otherwise a collision in the two nonzero signed classes modulo 5 yields a sum or difference divisible by 5 and by 2. The final two even inputs have product divisible by 4. Hence $18 \cdot 10 \cdot 4 = 720$.

C.2. **Six odd inputs.** Construct an 18-block as follows. If at least two inputs are divisible by 9, their sum or difference is divisible by 18. If exactly one is divisible by 9, the other five occupy four nonzero signed classes modulo 9; if none is divisible by 9, all six occupy those four classes. In both cases a collision gives a sum or difference divisible by 9 and by 2.

Among the remaining four odd inputs, construct a 10-block. If at least two are divisible by 5, take their sum or difference. If exactly one is divisible by 5, the other three occupy two nonzero signed classes; if none is divisible by 5, all four occupy those two classes. Thus a collision gives a sum or difference divisible by 10. The final odd pair yields 4. Therefore $18 \cdot 10 \cdot 4 = 720$.

C.3. **One odd and five even inputs.** If one of the five even inputs is divisible by 5, it supplies 10; reserve another even input as a factor 2, and construct a 36-multiple from the remaining four leaves. This gives $10 \cdot 2 \cdot 36 = 720$.

Otherwise the five even inputs lie in the four nonzero even signed classes modulo 20,

$$\{\pm2\}, \quad \{\pm4\}, \quad \{\pm6\}, \quad \{\pm8\}.$$

A collision gives a 20-block, while the other four leaves give 36. Thus $20 \cdot 36 = 720$.

C.4. **Five odd and one even input.** Write the odd inputs as a, b, c, d, f and the even input as e. If $5 \mid a$, among the other four odd inputs choose b, c with $12 \mid b \pm c$. Then $60 \mid a(b \pm c)$, while d, f, e yield a 12-multiple. Thus $60 \cdot 12 = 720$.

If no odd input is divisible by 5, the five odd inputs occupy four nonzero odd signed classes modulo 20, so a pair yields 20. The remaining three odd inputs yield 18, and multiplication by the unused even input turns that block into a multiple of 36. Thus $20 \cdot 36 = 720$.

C.5. **Two odd and four even inputs.** Let the odd inputs be a, b and the even inputs e_1, e_2, e_3, e_4.

A. A multiple of 5 occurs. If an odd input a is divisible by 5, choose three even inputs and construct a 48-multiple. The remaining odd-even pair yields a multiple of 3. Hence $5 \cdot 48 \cdot 3 = 720$.

If no odd input is divisible by 5 but an even input is, that even input supplies 10. From the remaining five leaves, a 2O1E triple yields 12 and the final even pair yields 6. Thus $10 \cdot 12 \cdot 6 = 720$.

Assume now that all six inputs are nonzero modulo 5.

B. The two odd inputs occupy one signed class modulo 5. Their suitable sum or difference is divisible by 10, while the four even inputs yield 72. Thus $10 \cdot 72 = 720$.

Henceforth the two odd inputs occupy different signed classes modulo 5.

C. Two even inputs occupy one signed class modulo 20. Their suitable sum or difference is divisible by 20, and the other four leaves yield 36. Thus $20 \cdot 36 = 720$.

We may therefore assume that the four even inputs occupy the four distinct signed classes

$$\{\pm2\}, \ \{\pm4\}, \ \{\pm6\}, \ \{\pm8\} \pmod{20}.$$

Group $\{\pm2\}, \{\pm8\}$ as X and $\{\pm4\}, \{\pm6\}$ as Y. These are the two nonzero signed classes modulo 5, and two even inputs lie in each group.

D. A multiple of 3 occurs. If an odd input a is divisible by 3, the other odd input b has the same modulo-5 signed class as two even inputs. Choose one of them, e, with a sign satisfying $5 \mid b \pm e$. The remaining three even inputs yield 48, so $3 \cdot 5 \cdot 48 = 720$.

If neither odd input is divisible by 3 but an even input e is, then e supplies 6. The two even inputs in the modulo-5 group opposite to e yield 10, and the remaining two odd inputs and one even input yield 12. Thus $6 \cdot 10 \cdot 12 = 720$.

Assume finally that all six inputs are coprime to 15.

E. No input is divisible by 3 or 5. The four even inputs occupy the three nonzero signed classes $\{\pm 1\}, \{\pm 2\}, \{\pm 4\}$ modulo 9, so two yield an 18-block. If this pair lies wholly in X or wholly in Y, the remaining two even inputs lie in the other modulo-5 class and yield 10, while the odd pair yields 4. This gives $18 \cdot 10 \cdot 4 = 720$.

If the 18-pair contains one input from each of X, Y, then the two unused even inputs also contain one from each group. Together with the two odd inputs, which occupy the two different modulo-5 classes, they form a 2O2E block and yield 40. Hence $18 \cdot 40 = 720$.

C.6. **Three odd and three even inputs.** Let a, b, c be odd and x, y, z even. If one even input is divisible by 3, it supplies 6, and the remaining five inputs yield 120. Otherwise the three even inputs are nonzero modulo 3 and occupy only the two residue classes $0, 2 \pmod 4$. Two, say x, y, are congruent modulo 4. Choose a sign with $3 \mid x \pm y$; the same sum or difference is divisible by 4, hence by 12. The remaining three odd inputs and one even input yield 60. Thus $12 \cdot 60 = 720$.

C.7. **Four odd and two even inputs.** Let a, b, c, d be odd and x, y even. If one even input is divisible by 3, it supplies 6 and the remaining five inputs yield 120.

Assume $3 \nmid xy$. If $x \equiv y \pmod 4$, choose the sign for which $3 \mid x \pm y$; the same block is divisible by 4, hence by 12. The four odd inputs yield 60, giving $12 \cdot 60 = 720$.

It remains that the two even inputs have different residues modulo 4. Relabel so that $4 \mid x$ and $y \equiv 2 \pmod 4$. If an odd input a is divisible by 3, then $12 \mid ax$, and the remaining three odd inputs and one even input yield 60.

Suppose no odd input is divisible by 3, but one odd input a is divisible by 5. Then $20 \mid ax$, while the remaining four inputs yield 36.

Finally assume all four odd inputs are coprime to 15. They yield 120 by Theorem 5.2. Since x, y are even and nonzero modulo 3, their suitable sum or difference is divisible by 6. Thus $120 \cdot 6 = 720$.

The seven subsections exhaust all parity patterns.

Appendix D. Parity-pattern guide to the factorial constructions

For quick reference, the table below lists every odd/even pattern through six inputs and points to a proved construction for the factorial target. The "additional block" column records parity-specific strengthenings that are explicitly used or singled out in the manuscript. Universal theorems, such as the further four-input guarantees of Section F, apply to every parity row at that input size and are therefore not repeated in every cell.

n	odd	parity pattern	factorial construction	additional displayed block(s) / proof source
1	0	0O1E	multiple of 1	the even singleton also gives 2; Section A
1	1	1O0E	multiple of 1	Section A
2	0	0O2E	multiple of 2	also 6, 8; Section A
2	1	1O1E	multiple of 2	universal 3 also holds; Section A
2	2	2O0E	multiple of 2	also 3, 4; Section A
3	0	0O3E	12 (hence 3!)	also 14, 18, 48; Theorems A.5, A.6 and A.8

n	odd	parity pattern	factorial construction	additional displayed block(s) / proof source
3	1	1O2E	12 (hence 3!)	Theorem A.5
3	2	2O1E	12 (hence 3!)	Theorem A.5
3	3	3O0E	12 (hence 3!)	also 14, 18; Theorems A.5 and A.6
4	0	0O4E	$24 = 4!$	also 48, 72; Theorems A.9, A.10 and 4.3
4	1	1O3E	$24 = 4!$	also 40, 48; Theorems A.9, A.12 and 4.3
4	2	2O2E	$24 = 4!$	also 40, 48; Theorems A.9, A.13 and 4.3
4	3	3O1E	$24 = 4!$	also 60; Theorems 4.3 and 5.1
4	4	4O0E	$24 = 4!$	also 48, 60; Theorems A.9, A.11 and 4.3
5	0	0O5E	$120 = 5!$	also 480; Theorems A.15 and 4.5
5	1	1O4E	$120 = 5!$	Theorem 4.5
5	2	2O3E	$120 = 5!$	Theorem 4.5
5	3	3O2E	$120 = 5!$	Theorem 4.5
5	4	4O1E	$120 = 5!$	Theorem 4.5
5	5	5O0E	$120 = 5!$	also 180, 240; Theorems A.14 and 4.5
6	0	0O6E	$18 \cdot 10 \cdot 4 = 720$	Section C.1
6	1	1O5E	$20 \cdot 36$ or $10 \cdot 2 \cdot 36$	Section C.3
6	2	2O4E	720 via the exhaustive branches in the proof	Section C.5
6	3	3O3E	$6 \cdot 120$ or $12 \cdot 60$	Section C.6
6	4	4O2E	720 via the exhaustive branches in the proof	Section C.7
6	5	5O1E	$60 \cdot 12$ or $20 \cdot 36$	Section C.4
6	6	6O0E	$18 \cdot 10 \cdot 4 = 720$	Section C.2

For $n = 5$, the single universal congruence proof of Theorem 4.5 covers all six parity rows at once. For $n = 6$, the seven rows correspond exactly to the seven exhaustive parity branches proved in Section C.

Appendix E. Provenance and dominance reductions for the recorded source-pool frontiers

Each modulus below is a separate existence guarantee. Products are formed only from disjoint input blocks. After the direct moduli and the available split products have been collected, a modulus m is deleted when another proved modulus M in the same class satisfies $m \mid M$. This is a genuine dominance deletion: the expression producing an M-multiple is itself an m-multiple. It is not an operation that combines two different expressions. For instance, 21 is deleted once 63 is proved because $21 \mid 63$, whereas separate guarantees 6 and 8 cannot be merged into a 24-guarantee without a separate proof.

The source pools used here consist of the direct theorems selected for this summary together with products of the earlier displayed source-pool frontiers over disjoint splits. The lists below give the retained values and the corresponding dominance deletions within those pools. They are not intended to be an exhaustive closure under every parity case split available elsewhere in the manuscript, and no optimality claim is made.

E.1. **One and two inputs.** For one input,

$$F_1 = O_1 = \{1\}, \qquad E_1 = \{2\}.$$

For two inputs, the universal guarantees are 2 and 3. In the all-odd class the additional modulus 4 dominates 2; in the all-even class the moduli 6 and 8 dominate 2 and 3 as indicated below:

$$F_2 = \{2, 3\}, \qquad O_2 = \{3, 4\}, \qquad E_2 = \{6, 8\}.$$

deleted modulus	retained dominator
$O_2:\ 2$	4
$E_2:\ 2$	6
$E_2:\ 3$	6

E.2. **Three inputs.** The universal direct constructions give $7, 8, 9, 10, 12$; the smaller split products $2, 3$ are dominated. The parity-restricted direct improvements are $14, 18$ for odd inputs and $14, 18, 48$ for even inputs. Thus

$$F_3 = \{7, 8, 9, 10, 12\},$$
$$O_3 = \{8, 10, 12, 14, 18\},$$
$$E_3 = \{10, 14, 18, 48\}.$$

The deletions are

F_3		O_3		E_3	
2	8	3	18	7	14
3	9	4	8	8	48
		7	14	9	18
		9	18	12	48
			16	48	

where each left entry divides the adjacent retained entry.

E.3. **Four inputs.** The universal source pool contains the one-plus-three and two-plus-two split products, together with the direct moduli $10, 14, 16, 24, 36$. After dominance reduction,

$$F_4 = \{10, 14, 16, 24, 36\}.$$

For four odd inputs the direct moduli 48 and 60 supplement the inherited and split products, giving

$$O_4 = \{14, 36, 48, 60\}.$$

For four even inputs the retained products and direct theorem are

$$20 = 2 \cdot 10, \quad 28 = 2 \cdot 14, \quad 64 = 8 \cdot 8, \quad 72, \quad 96 = 2 \cdot 48,$$

so

$$E_4 = \{20, 28, 64, 72, 96\}.$$

The complete deletions from these source pools are

F_4		O_4		E_4	
4	16	8	48	36	72
6	24	9	36	48	96
7	14	10	60		
8	16	12	48		
9	36	16	48		
12	24	18	36		

E.4. **Five inputs.** A disjoint $3+2$ split gives the universal candidates

$$14, 16, 18, 20, 21, 24, 27, 30, 36,$$

and the direct factorial theorem adds 120. Since

$$18 \mid 36, \qquad 20 \mid 120, \qquad 24 \mid 120, \qquad 30 \mid 120,$$

the universal frontier is

$$F_5 = \{14, 16, 21, 27, 36, 120\}.$$

For five odd inputs, the $3\mathrm{O} + 2\mathrm{O}$ products give

$$24, 30, 32, 36, 40, 42, 48, 54, 56, 72,$$

and the direct theorems add $180, 240$. The retained values are

$$O_5 = \{32, 42, 54, 56, 72, 180, 240\},$$

with deletions

$$24 \mid 240, \quad 30 \mid 180, \quad 36 \mid 180, \quad 40 \mid 240, \quad 48 \mid 240.$$

For five even inputs, the $3\mathrm{E} + 2\mathrm{E}$ products give

$$60, 80, 84, 108, 112, 144, 288, 384,$$

and the direct theorem adds 480. Therefore

$$E_5 = \{84, 108, 112, 288, 384, 480\},$$

with

$$60 \mid 480, \qquad 80 \mid 480, \qquad 144 \mid 288.$$

E.5. **Six inputs.** For arbitrary inputs, the retained $3+3$ products are

$$49 = 7 \cdot 7, \quad 56 = 7 \cdot 8, \quad 63 = 7 \cdot 9, \quad 64 = 8 \cdot 8, \quad 70 = 7 \cdot 10,$$
$$81 = 9 \cdot 9, \quad 84 = 7 \cdot 12, \quad 96 = 8 \cdot 12, \quad 100 = 10 \cdot 10, \quad 108 = 9 \cdot 12.$$

The direct six-input theorem adds 720, giving

$$F_6 = \{49, 56, 63, 64, 70, 81, 84, 96, 100, 108, 720\}.$$

All other candidates produced by the $1+5$, $2+4$, and $3+3$ splits are removed by the following explicit divisibilities:

14	56	16	64	20	100	21	63
27	81	28	56	30	720	32	64
36	108	42	84	48	96	72	720
80	720	90	720	120	720	144	720

For six odd inputs, the retained products are

$$100 = 10 \cdot 10, \quad 112 = 8 \cdot 14, \quad 140 = 10 \cdot 14, \quad 168 = 12 \cdot 14, \quad 192 = 4 \cdot 48,$$
$$196 = 14 \cdot 14, \quad 216 = 12 \cdot 18, \quad 252 = 14 \cdot 18, \quad 324 = 18 \cdot 18,$$

together with 720. Thus

$$O_6 = \{100, 112, 140, 168, 192, 196, 216, 252, 324, 720\}.$$

The deleted candidates and one retained dominator for each are

32	192	42	168	54	216
56	112	64	192	72	216
80	720	96	192	108	216
120	720	144	720	180	720
240	720				

For six even inputs, the retained products are

$$100 = 10 \cdot 10, \quad 140 = 10 \cdot 14, \quad 196 = 14 \cdot 14, \quad 252 = 14 \cdot 18, \quad 324 = 18 \cdot 18,$$
$$512 = 8 \cdot 64, \quad 672 = 14 \cdot 48, \quad 864 = 18 \cdot 48, \quad 960 = 2 \cdot 480, \quad 2304 = 48 \cdot 48,$$

together with 720. Hence

$$E_6 = \{100, 140, 196, 252, 324, 512, 672, 720, 864, 960, 2304\}.$$

The complete deletion list is

120	720	160	960	168	672
180	720	216	864	224	672
384	2304	432	864	480	960
576	2304	768	2304		

These source and deletion lists justify the displayed source-pool frontiers. They do not exclude further nondominated moduli obtainable from other proved case splits or from later strengthening arguments.

Appendix F. Further universal four-input guarantees

This appendix records additional universal four-input divisibility guarantees beyond the source pool used to organize the factorial-through-six proof. Each statement below is a separate existence theorem for a legal read-once expression using only $+, -, \times$ and every input exactly once.

F.1. **Two auxiliary three-even blocks.** The following four-input refinements use two convenient three-even strengthenings.

Lemma F.1 (Three even inputs give 20)**.** *Any three even integers admit a read-once expression divisible by* 20*.*

Proof. Let the inputs be a, b, c. If one is divisible by 5, their product is already divisible by 40, hence by 20. Otherwise the three nonzero residues modulo 5 occupy the two signed classes $\{\pm 1\}, \{\pm 2\}$. Two inputs, say a, b, lie in one class, so one of $a + b, a - b$ is divisible by 5. It is also even, and the remaining input c is even; hence $(a \pm b)c$ is divisible by 20. □

Lemma F.2 (Three even inputs give 32)**.** *Any three even integers admit a read-once expression divisible by* 32*.*

Proof. Let the inputs be a, b, c. If $32 \mid abc$, use the product. Otherwise no input is divisible by 8, and two inputs cannot both be divisible by 4. If exactly one input, say c, is divisible by 4, then $a, b \equiv 2 \pmod 4$. Modulo 8 they lie in the single signed class $\{\pm 2\}$, so one of $a + b, a - b$ is divisible by 8; multiplying by c gives a multiple of 32.

It remains that none of the three inputs is divisible by 4. Then all are $2 \pmod 4$, and modulo 16 they occupy only the two signed classes $\{\pm 2\}$ and $\{\pm 6\}$. Two, say a, b, occupy one class, so one of $a + b, a - b$ is divisible by 16. Multiplication by the remaining even input gives a multiple of 32. □

F.2. **Signed-class preliminaries for four inputs.** For an odd modulus m, write $[r]_m$ for the signed residue class $\{r, -r\}$ modulo m. When all classes under discussion are nonzero, their products are defined by

$$[r]_m[s]_m = [rs]_m.$$

This notation is only a bookkeeping device for congruences; no division of input values is introduced. By Theorem 3.4, leaves may be oriented to convenient signed representatives and restored afterward without changing divisibility.

Lemma F.3 (Small signed-class relation)**.** *Let G be a finite abelian group with $|G| \le 6$. Any four-element subset contains three distinct elements X, Y, Z with $XY = Z$.*

Proof. Write G additively. If the identity is not in the four-element set A, fix $x \in A$. The three sums $x + y$ with $y \in A \setminus \{x\}$ are distinct. If none equals a third, distinct member of A, all three lie outside A, but there are at most two elements outside A, a contradiction. If the identity is in A, write $A = \{0, x, y, z\}$. The three sums $x + y, x + z, y + z$ are distinct, and the same argument applies. □

F.3. **The universal moduli 17 and 19.**

Theorem F.4. *The universal guarantees $P(4;17)$ and $P(4;19)$ hold.*

Proof. Fix $p \in \{17, 19\}$. If one input is divisible by p, multiply it by any legal expression on the other three leaves. If two inputs occupy the same nonzero signed class modulo p, their suitable sum or difference is a p-multiple and may be multiplied by the remaining two leaves. We may therefore assume that the four signed classes are nonzero and distinct.

First suppose none is the identity class $[1]_p$. Consider the four singleton classes together with their six pair-products. If no pair-product equals a third singleton class and no two disjoint pair-products are equal, then these ten classes are all distinct: equality of pair-products sharing a factor would force two singleton classes to be equal, while equality of a pair-product with one of its own factors would force the other factor to be $[1]_p$. But the numbers of nonzero signed classes are 8 for $p = 17$ and 9 for $p = 19$. Thus a useful relation is forced. A relation $[a]_p[b]_p = [c]_p$ gives $(ab \pm c)d$, and equality of disjoint pair-products gives $ab \pm cd$.

It remains that one class is $[1]_p$. The cases not already having one of the preceding multiplicative relations are exactly the rows in the following table. These rows do not enumerate integer input quadruples. They enumerate the residual signed residue-class configurations modulo p after the preceding general congruence reductions. Thus, for example, the row $(1, 2, 3, 4)$ for $p = 17$ means that, after sign normalization and relabelling,

$$(a, b, c, d) \equiv (1, 2, 3, 4) \pmod{17};$$

it does not mean that the original integers themselves are $1, 2, 3, 4$. The listed construction then gives

$$(a + b - c)d \equiv (1 + 2 - 3)4 \equiv 0 \pmod{17}.$$

Every integer quadruple is covered either by one of the preceding general relations or, after sign normalization and relabelling, by one of the rows below. Each row therefore represents an entire signed congruence class of integer inputs, not a numerical test case.

p	representatives (a, b, c, d)	expression
17	(1,2,3,4)	$(a + b - c)d$
17	(1,2,4,5)	$(a + b)c + d$
17	(1,2,4,6)	$ab + c - d$
17	(1,2,4,7)	$a + b + c - d$
17	(1,2,6,7)	$a - b - c + d$
17	(1,3,4,8)	$a + b + c - d$
17	(1,3,5,8)	$a + b + c + d$
17	(1,4,5,8)	$(a + b - c)d$
17	(1,4,6,8)	$a - b - c - d$
17	(1,4,7,8)	$c + (a - b)d$
19	(1,2,3,4)	$(a + b - c)d$
19	(1,2,3,5)	$(a + b - c)d$
19	(1,2,4,6)	$ab + c - d$
19	(1,2,4,7)	$a + b + c - d$
19	(1,2,5,6)	$a - b - c + d$
19	(1,2,6,8)	$ab + c - d$
19	(1,3,4,8)	$a + b + c - d$
19	(1,3,4,9)	$(a + b - c)d$
19	(1,3,5,9)	$a + b + c - d$
19	(1,3,7,9)	$(a + b)c - d$
19	(1,4,6,9)	$a - b - c + d$
19	(1,5,6,7)	$a + b + c + d$
19	(1,5,6,9)	$(a + b - c)d$
19	(1,5,8,9)	$(a + b)c + d$

The list is exhaustive because the identity-containing four-subsets of the signed-class multiplication table were reduced by the same two multiplicative relations used above. Direct substitution verifies every row, and sign transport returns a legal expression on the original inputs. □

F.4. **The universal modulus 21.**

Theorem F.5. *The universal guarantee $P(4;21)$ holds.*

Proof. If one input is divisible by 7, retain its factor 7 and combine the remaining three leaves into a 3-multiple by first using Theorem 3.2 on two of them and multiplying by the third. Otherwise the four inputs occupy the three nonzero signed classes modulo 7, so a collision pair has a suitable sum or difference divisible by 7. The complementary pair gives a 3-multiple by Theorem 3.2. The two blocks are disjoint, hence their product is divisible by 21. □

F.5. **The universal modulus 22.**

Theorem F.6. *The universal guarantee $P(4;22)$ holds.*

Proof. If one input is divisible by 11, multiply it by an even legal expression on the other three leaves: use their product if one of those leaves is even, and use $(b-c)d$ if all three are odd. If instead two nonzero inputs collide in a signed class modulo 11, their suitable sum or difference is an 11-block; the complementary two leaves can always be combined into an even expression, using a sum or difference for equal parity and the product for opposite parity. Thus both situations give a 22-multiple.

Assume henceforth that the four signed classes modulo 11 are nonzero and distinct. The five nonzero signed classes form an abelian multiplication table of order five, so Theorem F.3 supplies a relation $[a][b] = [c]$. If the remaining leaf d is even, or if ab and c have the same parity, the appropriate $(ab \pm c)d$ is already even and divisible by 11. A direct parity check of the possible relations shows that only the pattern 3O1E can remain.

Let u, x, y be the three odd leaves and e the even leaf. If $[u][x] = [y]$ for some relabelling, then $e(ux \pm y)$ gives 22. Otherwise the relation-free triples of distinct signed representatives modulo 11 are

$$\{1,2,3\}, \quad \{1,2,4\}, \quad \{1,3,5\}, \quad \{1,4,5\}.$$

Over all relabellings of the three odd leaves, the even-valued forms $ux \pm y$ and $u(x \pm y)$ attain every possible distinct external signed class except one case:

$$\{[u],[x],[y]\} = \{[1],[2],[4]\}, \qquad [e] = [3].$$

For this last case orient the representatives as $u = 1, x = 2, y = 4, e = 3$. Then

$$u - x + ye = 1 - 2 + 4 \cdot 3 = 11,$$

and the expression is even. Thus every branch gives a multiple of 22. □

F.6. **The universal modulus 23.**

Theorem F.7. *The universal guarantee $P(4;23)$ holds.*

Proof. A leaf divisible by 23, or a collision of two nonzero signed classes modulo 23, immediately gives a 23-multiple using all four leaves. We may therefore assume four distinct nonzero signed classes.

Suppose first that the identity class [1] is absent. Assume, toward a contradiction, that there is no relation $[a][b] = [c]$, no equality of disjoint pair-products, and no relation in which the product of three classes equals the fourth. The four singleton classes and their six pair-products are then ten distinct classes. Indeed, two pair-products that share a factor cannot be equal, a pair-product cannot equal a third singleton by assumption, and it cannot equal one of its own factors because [1] is absent. There are eleven nonzero signed classes modulo 23, so at most one class is not among these ten.

Now consider the four triple-products. They are distinct, so at least three of them lie among the ten classes just described. A triple-product cannot equal a pair-product without either

cancelling a common factor to obtain a forbidden pair-to-singleton relation or forcing an identity singleton. If a triple-product equals a singleton different from the omitted member, then cancelling the shared singleton shows that the other two classes form an identity-product pair. One such pair accounts for at most two of the four triple-products. Hence at least two identity-product pairs are forced. They must be disjoint, since the four classes are distinct and nonidentity; their two pair-products are both [1], contradicting the assumed absence of an equality of disjoint pair-products. Thus one of the useful relations exists, and it gives a legal expression of the form $(ab \pm c)d$, $ab \pm cd$, or $abc \pm d$.

It remains that one class is [1]. Orient and order the signed representatives as

$$a = 1, \qquad 2 \le x < y < z \le 11.$$

For two representatives put

$$C(r, s) = \{[rs], [r + s], [r - s]\}_{23}.$$

If, for one of the three pairings,

$$C(1, x) \cap C(y, z), \quad C(1, y) \cap C(x, z), \quad C(1, z) \cap C(x, y)$$

is nonempty, two legal two-leaf expressions occupy one signed class and their suitable sum or difference gives a four-leaf multiple of 23. After also removing the pair-to-singleton and disjoint pair-product relations used above, the complete residual list is

x	y	z	legal 23-multiple
2	3	9	$(a + x - y)z$
3	5	11	$x(a - y) - z$
4	5	11	$(a + x - y)z$
4	7	8	$x(a + y - z)$
4	7	9	$(a + y)x - z$
6	7	8	$(a + x - y)z$
6	8	11	$(a + x)z - y$
8	9	10	$(a - z + y)x$
9	10	11	$(a - z + y)x$

For $x = 5, 7, 10, 11$ no residual pair (y, z) remains. The displayed table is obtained directly from the eleven signed classes and the three displayed C-intersections, so it exhausts the identity branch. Each last-column expression is visibly 0 modulo 23. This completes the proof. □

F.7. **The universal modulus** 25**.**

Theorem F.8. *The universal guarantee* $P(4; 25)$ *holds.*

Proof. If one input is divisible by 5, retain that factor and use a three-leaf 5-multiple on the other inputs, available because $P(3; 10)$ is already proved. Thus suppose all four leaves are units modulo 5. There are two nonzero signed classes, $\{\pm 1\}$ and $\{\pm 2\}$.

If their multiplicities are $4 + 0$ or $2 + 2$, partition the leaves into two same-class pairs. Each pair has a suitable sum or difference divisible by 5, and the product of the two disjoint blocks is divisible by 25.

It remains to handle a $3 + 1$ distribution. Both orientations must be considered.

Three $\{\pm 1\}$ *leaves and one* $\{\pm 2\}$ *leaf.* Orient

$$x = 1 + 5\alpha, \quad y = 1 + 5\beta, \quad z = 1 + 5\gamma, \quad w = 2 + 5\delta \pmod{25},$$

with $\alpha, \beta, \gamma, \delta$ residues modulo 5. If two of α, β, γ coincide, the corresponding difference is already a 25-block. Otherwise put

$$A = \{\alpha, \beta, \gamma\}, \qquad S = \alpha + \beta + \gamma, \qquad r = \delta - S.$$

The following legal expressions are divisible by 25 under the indicated conditions:

condition on r	expression
$r = 0$	$xy + z - w$
$r = \alpha$ (cyclically)	$x(y + z) - w$
$r = 2\alpha$ (cyclically)	$y + z - xw$
$r = -2\beta$ (cyclically)	$(w - x)y - z$

Indeed, expansion modulo 25 reduces these four rows respectively to multiples of 5 with coefficients $-r$, $\alpha - r$, $-3\alpha - r$, and $r + 2\beta$ modulo 5.

It remains only to see that these conditions cover every r. If $0 \notin A$, the three elements of A are three of the four nonzero residues. Let t be the missing nonzero residue. Then $3t$ is nonzero and different from t, so $3t \in A$; doubling gives $2(3t) = t$ modulo 5, hence $t \in 2A$. If $0 \in A$, write the two nonzero members as u and v. The third nonzero direction v is one of $-u, 2u, -2u$. If $v = -u$, then $A \cup 2A$ already contains $u, -u, 2u, -2u$; in the other two cases the missing nonzero direction is supplied by $2A$ or $-2A$. Hence

$$\{0\} \cup A \cup 2A \cup (-2A)$$

is the full residue system modulo 5.

Three $\{\pm 2\}$ *leaves and one* $\{\pm 1\}$ *leaf.* Orient

$$x = 2 + 5\alpha, \quad y = 2 + 5\beta, \quad z = 2 + 5\gamma, \quad w = 1 + 5\delta \pmod{25}.$$

Again a repeated lift among α, β, γ gives a 25-difference. Assume they are distinct and use the same S and $r = \delta - S$. The family

$$x(y - w) - z$$

has condition $r = 2\alpha + \gamma = S + (\alpha - \beta)$ after relabelling, so as the ordered pair α, β varies it covers every residue except S. The family

$$z(wx - y)$$

has condition $r = -S + 2(\alpha - \beta)$ and therefore covers every residue except $-S$. The ordered differences of any three-element subset modulo 5 are all four nonzero residues. If $S \neq 0$, the two families therefore cover all r.

If $S = 0$, both families miss only $r = 0$. The three lifts are then either $\{0, 1, -1\}$ or $\{0, 2, -2\}$. In the first case orient $\gamma = 1$ and use $z(w + x + y)$; in the second orient $\gamma = -2$ and use $z(w + xy)$. The inner parenthesis is a multiple of 25 in each case. Thus every $3 + 1$ branch is covered, proving $P(4; 25)$. □

F.8. **The universal modulus** 26**.**

Theorem F.9. *The universal guarantee* $P(4; 26)$ *holds.*

Proof. If one input is divisible by 13, multiply it by an even legal expression on the other three leaves. If two nonzero inputs collide in one signed class modulo 13, their suitable sum or difference is a 13-block and the complementary pair can be combined into an even expression. Hence assume four distinct nonzero signed classes. By Theorem F.3, some three distinct classes satisfy $[a][b] = [c]$. The construction $(ab \pm c)d$ is a 26-multiple whenever the bracket is even or d is even. A complete parity check leaves only 2O2E and 3O1E residuals.

For 2O2E, orient the odd leaves as p, q and the even leaves as x, y. After all parity-compatible pair-to-third relations have been used, exactly six signed-class configurations remain. The last column gives an even expression that is 0 modulo 13.

ordered odd representatives (p, q)	ordered even representatives (x, y)	expression
$(1, 2)$	$(3, 4)$	$(p + q - x)y$
$(1, 5)$	$(2, 4)$	$x + (p + q)y$
$(1, 5)$	$(2, 6)$	$x(p + q - y)$
$(1, 5)$	$(3, 4)$	$x + (p - q)y$
$(1, 5)$	$(3, 6)$	$x(p + q - y)$
$(1, 6)$	$(3, 4)$	$p + q - x - y$

For 3O1E, let u, x, y be odd and e even. Besides the direct pair-to-third relations, use the even three-odd forms $xy \pm u$ and $x(y \pm u)$. If the signed class of one of these forms equals $[e]$, the suitable sum or difference with e is both even and divisible by 13. The complete signed-class reduction leaves only

odd classes	$[e]$	expression
$\{1, 2, 4\}$	5	$u - (x - e)y$
$\{1, 3, 5\}$	4	$u - x(y + e)$
$\{3, 4, 6\}$	2	$u(x - y + e)$

with representatives in the displayed order. Their values are respectively $13, -26, 0$ modulo 13, and all three expressions are even. Thus every residual case gives a 26-multiple. □

F.9. **The universal modulus** 27**.**

Theorem F.10. *The universal guarantee* $P(4; 27)$ *holds.*

Proof. If one input is divisible by 3, retain its factor 3 and use the three-input 9 theorem Theorem A.3 on the remaining leaves. Otherwise the four inputs are units modulo 9, whose nonzero signed classes are $\{\pm 1\}, \{\pm 2\}, \{\pm 4\}$. Two inputs collide in one class, producing a 9-block; the complementary pair gives a 3-multiple by Theorem 3.2. The disjoint product is divisible by 27. □

F.10. **The universal modulus** 28**.**

Theorem F.11. *The universal guarantee* $P(4; 28)$ *holds.*

Proof. If an input is divisible by 7, multiply it by an 8-multiple from the remaining three inputs, available by Theorem A.2; the result is in fact a multiple of 56. Suppose no input is divisible by 7. Then four inputs occupy the three nonzero signed classes modulo 7, so a collision exists.

If a collision pair has the same parity, its suitable sum or difference is divisible by both 7 and 2, hence is a 14-block. The complementary pair can be made even, giving 28.

Assume therefore that every collision is odd-even. For 3O1E, the three odd inputs give a 14-multiple and multiplication by the even leaf gives 28. For 1O3E, if one even input is divisible by 4, retain that factor 4 and use the three-input 7 theorem on the other leaves. If not, all three even inputs are 2 (mod 4). Since no even-even collision is allowed, they occupy all three signed classes modulo 7. The odd input collides with one even input, giving a 7-block; the product of the two remaining even inputs supplies a factor 4.

It remains to consider 2O2E. If one even input is divisible by 4, argue as above. Otherwise the two even inputs x, y are both 2 (mod 4). The two odd signed classes and the two even signed classes are two-element subsets of the three nonzero signed classes modulo 7. If the two subsets are equal, one cross pairing has matched classes, so the corresponding two odd-even products occupy one signed class. If the subsets differ, they share one class C; call the other two classes D, E. Direct multiplication in the three-class table gives

$$CC = DE.$$

Thus in either case one of the two cross pairings produces products in the same signed class modulo 7. Their suitable sum or difference is divisible by 7. Each odd-even product is 2 (mod 4), so the same sum or difference is divisible by 4. Hence it is a 28-multiple. The all-odd and all-even patterns cannot enter the residual branch because any collision there has the same parity. □

F.11. **The universal modulus** 30**.**

Theorem F.12. *The universal guarantee* $P(4; 30)$ *holds.*

Proof. If one input is divisible by 3, retain that factor and use the three-input 10 theorem Theorem A.4. If one input is divisible by 5, retain that factor and use the three-input 12 theorem Theorem A.5; this gives a multiple of 60. Hence assume all four inputs are coprime to 15.

Modulo 5 there are two nonzero signed classes. If one class contains a same-parity pair, that pair has a suitable sum or difference divisible by both 5 and 2, hence a 10-block; the complementary pair gives a 3-multiple. The only residual parity pattern is therefore 2O2E, with each modulo-5 signed class containing one odd and one even input. Orient the classes so that $a, x \equiv 1 \pmod 5$ and $b, y \equiv 2 \pmod 5$, where a, b are odd and x, y are even. Consider

$$E_1 = ax + sby, \qquad E_2 = ay + sbx, \qquad s \in \{\pm 1\}.$$

Modulo 5, E_1 requires $s = +1$ while E_2 requires $s = -1$. Modulo 3, orient the four unit residues as $\alpha, \beta, \xi, \eta \in \{\pm 1\}$. The two pairs of product terms have the same product

$$(\alpha\xi)(\beta\eta) = (\alpha\eta)(\beta\xi),$$

so the sign required to cancel the two terms modulo 3 is the same for E_1 and E_2. If that common sign is $+1$, use E_1; if it is -1, use E_2. The selected expression is divisible by both 3 and 5, and it is even because each product term is odd-even. Hence it is divisible by 30. □

F.12. **A complementary binary lift.**

Lemma F.13 (Binary complementary lift)**.** *Suppose two legal read-once expressions satisfy*

$$E_0 \equiv ML \pmod{2M}, \qquad E_1 \equiv M(L+1) \pmod{2M}$$

for an integer L. Then one of E_0, E_1 is divisible by $2M$.

Proof. The consecutive integers L and $L+1$ have opposite parity. □

F.13. **The universal modulus 32.**

Theorem F.14. *The universal guarantee $P(4; 32)$ holds.*

Proof. The parity patterns other than 3O1E follow from the smaller blocks already proved: four odd inputs split into an 8-pair and a 4-pair; 2O2E splits into a 4-odd pair and an 8-even pair; and in the 1O3E and 4E patterns, three even leaves give 32 by Theorem F.2, after which the remaining leaf is multiplied in.

Let a, b, c be odd and e even. If $4 \mid e$, the three odd leaves give an 8-multiple and multiplication by e gives 32. Hence assume $e \equiv 2 \pmod 4$. If two odd leaves occupy one signed class modulo 16, their suitable sum or difference is a 16-block, and multiplication by the other odd leaf and e gives 32. If three distinct odd classes satisfy $[a][b] = [c]$ modulo 16, use $e(ab \pm c)$.

The odd signed classes modulo 16 are represented by $1, 3, 5, 7$. Direct multiplication shows that the only relation-free triples are

$$R_A = \{1, 3, 7\}, \qquad R_B = \{1, 5, 7\}.$$

The signed class of e modulo 16 is represented by 2 or 6. Orient the leaves accordingly and write, modulo 32,

$$a = r_a + 16A, \quad b = r_b + 16B, \quad c = r_c + 16C, \quad e = s + 16D,$$

where $A, B, C, D \in \{0, 1\}$ and $s \in \{2, 6\}$. Direct expansion gives the following complementary pairs; every congruence is modulo 32.

odd classes	s	E_0	E_1
$(1,3,7)$	2	$c(a-b+e) \equiv 16(A+B+D)$	$ce+b-a \equiv 16(1+A+B+D)$
$(1,3,7)$	6	$c(e+b)+a \equiv 16(A+B+C+D)$	$a(e+c)+b \equiv 16(1+A+B+C+D)$
$(1,5,7)$	2	$ab+e-c \equiv 16(A+B+C+D)$	$a-c(b+e) \equiv 16(1+A+B+C+D)$
$(1,5,7)$	6	$c(a+b-e) \equiv 16(A+B+D)$	$a+b+ce \equiv 16(1+A+B+D)$

Each row has the form of Theorem F.13 with $M = 16$, so one displayed expression is divisible by 32. This covers the final parity pattern. □

F.14. **The universal modulus 33.**

Theorem F.15. *The universal guarantee $P(4;33)$ holds.*

Proof. If one input is divisible by 11, multiply it by a three-leaf 3-multiple. If two nonzero inputs collide in one signed class modulo 11, their suitable sum or difference is an 11-block and the complementary pair gives a 3-multiple. Thus assume four distinct nonzero signed classes modulo 11. Orient them to one of the five representative sets

$$1234, \quad 1235, \quad 1245, \quad 1345, \quad 2345.$$

For later use, the following expressions are all 0 modulo 11 in their respective rows.

classes (a,b,c,d)	standard 11-relations
1234	$a(bd+c),\ b(cd-a),\ ac+bd$
1235	$a(bc+d),\ c(bd+a),\ ad+bc$
1245	$c(bd+a),\ a(cd+b),\ ab+cd$
1345	$d(bc-a),\ a(bd-c),\ ac-bd$
2345	$c(ab+d),\ d(ac+b),\ a(bd-c),\ b(cd+a),\ ad+bc$

We now impose divisibility by 3 on the same expression. If at least two leaves are 0 modulo 3, direct inspection of the displayed standard relations gives one in which every surviving term is also 0 modulo 3. Suppose exactly one leaf is 0 modulo 3. The following table gives an 11-multiple on the other three leaves whenever one exists; multiplying that three-leaf expression by the 3-divisible leaf immediately gives 33. A dash marks the only four hard positions.

classes	$a\equiv 0$	$b\equiv 0$	$c\equiv 0$	$d\equiv 0$
1234	$bd+c$	$a-d+c$	–	$a-c+b$
1235	$b-d+c$	–	$a+bd$	$a-c+b$
1245	$b+d+c$	$a-d+c$	$a+bd$	–
1345	$bd-c$	$a-d+c$	–	$a-c+b$
2345	$bd-c$	$a+d+c$	$a-d+b$	$ac+b$

In a hard position the remaining three residues are nonzero modulo 3; the two subcases in the next table cover them because two nonzero residues modulo 3 are either equal or opposite.

classes	3-divisible leaf	repair
1234	c	if $a\equiv b$, use $a+cd-b$; otherwise use $(a+b-c)d$
1235	b	if $a\equiv c$, use $(a+b-c)d$; otherwise use $(a+c)d+b$
1245	d	if $a\equiv c$, use $(a-c)b-d$; otherwise use $(a+c-d)b$
1345	c	if $a\equiv b$, use $cd-a+b$; otherwise use $(a+b-c)d$

Every repair is 0 modulo 11 and, by the stated modulo-3 condition, also 0 modulo 3.

Finally assume all four leaves are units modulo 3. After changing the overall sign, normalize the modulo-3 residue of a to 1. There are exactly eight modulo-3 residue-sign triples for (b,c,d). Applying the standard relations to these finite congruence possibilities leaves only the following residual patterns; the last column gives a repair that is simultaneously 0 modulo 11 and modulo 3.

classes	$(a,b,c,d)\pmod 3$	repair
1234	$(1,-1,-1,1)$	$a+bc+d$
1234	$(1,-1,1,-1)$	$a+d-b-c$
1235	$(1,-1,1,-1)$	$a+b+c+d$
1235	$(1,1,1,1)$	$(a+c)d+b$
1245	$(1,-1,1,-1)$	$a+d-b-c$
1245	$(1,1,1,1)$	$a+d-b-c$
1345	$(1,1,-1,1)$	$a-b-c-d$
1345	$(1,-1,1,1)$	$a-b-c-d$

The class set 2345 has no unit residual. The zero, one-zero, and all-unit branches exhaust the modulo-3 possibilities, so $P(4;33)$ follows. □

F.15. **The universal modulus** 35**.**

Theorem F.16. *The universal guarantee* $P(4;35)$ *holds.*

Proof. If one input is divisible by 5, retain that factor and apply the three-input 7 theorem Theorem A.1 to the remaining leaves. If one input is divisible by 7, retain that factor and use a three-leaf 5-multiple on the remaining leaves. Hence assume all four inputs are units modulo 35.

If two inputs occupy the same signed class modulo 35, their suitable sum or difference is a 35-block. If the four leaves can be split into two pairs so that one pair collides modulo 5 and the other pair collides modulo 7 (with the roles of 5 and 7 interchangeable), the product of the two suitable pair expressions is divisible by 35. We may also dispose of a relation $[a][b] = [c]$ by $(ab \pm c)d$, and of an equality of disjoint pair-products by $ab \pm cd$.

The unit signed classes modulo 35 have representatives

$$1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17.$$

After the preceding reductions, the complete residual list consists of the following 28 four-class sets. As above, this is a finite classification of signed residue classes modulo 35, not an enumeration of integer inputs. Each representative tuple denotes all integer quadruples that normalize to that signed congruence configuration. The expression in the right column is 0 modulo 35; sign transport restores the original leaves.

	representatives (a, b, c, d)	legal 35-multiple
1	(1,2,3,4)	$(a + b - c)d$
2	(1,2,3,9)	$(a + b - c)d$
3	(1,2,4,6)	$ab + c - d$
4	(1,2,4,11)	$b - (a - c)d$
5	(1,2,4,13)	$a + b(cd)$
6	(1,2,6,11)	$a - bc + d$
7	(1,2,8,11)	$a - b(cd)$
8	(1,3,6,9)	$ab + c - d$
9	(1,3,6,16)	$a + bc + d$
10	(1,3,8,16)	$a + b(cd)$
11	(1,3,9,13)	$a - b(cd)$
12	(1,3,9,16)	$a + b - cd$
13	(1,4,6,12)	$a - bc - d$
14	(1,4,8,12)	$a + b(cd)$
15	(1,4,11,12)	$b(a + c - d)$
16	(1,4,12,17)	$b(a + c) - d$
17	(1,6,9,17)	$a - bc - d$
18	(1,6,11,12)	$b(a + c - d)$
19	(1,6,16,17)	$b(a + c - d)$
20	(1,8,9,17)	$a + b(cd)$
21	(1,9,12,17)	$a + c - bd$
22	(1,9,16,17)	$a - b - cd$
23	(1,11,12,13)	$a - b(cd)$
24	(1,13,16,17)	$a - b(cd)$
25	(4,8,11,16)	$b - c(a - d)$
26	(4,11,13,16)	$b - a(c + d)$
27	(8,9,11,16)	$a(b - c) + d$
28	(9,11,13,16)	$ab - c - d$

The table is exhaustive at the signed-congruence level: it is the list of four-element subsets of the twelve displayed signed unit classes that remain after deleting exactly the four preceding kinds of relations. Thus no unit configuration is omitted, and the theorem follows. □

F.16. **The universal modulus** 40**.**

Theorem F.17. *The universal guarantee* $P(4;40)$ *holds.*

Proof. Four odd inputs give 40 by a modulo-5 collision: a collision pair is a 10-block and the remaining odd pair is a 4-block; if an odd input is itself divisible by 5, retain its factor 5 and use a three-odd 8-block. The patterns 2O2E and 1O3E are the earlier 40 theorems Theorems A.12

and A.13. Four even inputs are handled by choosing three even leaves for the 20-block of Theorem F.1 and multiplying by the fourth. Only 3O1E remains.

Let a, b, c be odd and e even. If $4 \mid e$, the three odd inputs give a 10-multiple and multiplication by e gives 40. If an odd input is divisible by 5, retain its factor 5 and use a three-input 8-block on the other leaves. If $5 \mid e$, then e supplies a factor 10 and the three odd leaves supply a factor 4. Hence assume $e \equiv 2 \pmod 4$ and no input is divisible by 5.

If two odd leaves occupy one signed class modulo 20, their suitable sum or difference is a 20-block and multiplication by the third odd leaf and e gives 40. If three distinct odd classes satisfy $[a][b] = [c]$, use $e(ab \pm c)$. The odd signed classes modulo 20 are represented by $1, 3, 7, 9$, and direct multiplication leaves only the relation-free triples

$$R_A = \{1, 3, 9\}, \qquad R_B = \{1, 7, 9\}.$$

Orient the even leaf to $s = 2$ or 6 modulo 20 and write, modulo 40,

$$a = r_a + 20A, \quad b = r_b + 20B, \quad c = r_c + 20C, \quad e = s + 20D,$$

where $A, B, C, D \in \{0, 1\}$. The residual lift table is

odd classes	s	E_0	E_1
$(1, 3, 9)$	2	$c(a - b + e) \equiv 20(A + B + D)$	$ce + b - a \equiv 20(1 + A + B + D)$
$(1, 3, 9)$	6	$ab + e - c \equiv 20(A + B + C + D)$	$bc - a - e \equiv 20(1 + A + B + C + D)$
$(1, 7, 9)$	2	$ab + e - c \equiv 20(A + B + C + D)$	$bc - a - e \equiv 20(1 + A + B + C + D)$
$(1, 7, 9)$	6	$c(a - b + e) \equiv 20(A + B + D)$	$ce + b - a \equiv 20(1 + A + B + D)$

All congruences are modulo 40. Each row is a binary complementary lift with $M = 20$, so one expression is divisible by 40. □

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